

PROFESSIONAL SUMMARY

AI infrastructure engineer with nearly five years of cloud-native AI/HPC experience, focused on LLM serving since 2024. Builds Kubernetes Operator-based model-serving control planes for heterogeneous accelerator fleets across service lifecycle, P/D topology and capacity planning, request routing, and failure recovery; also delivers production multi-tenant batch inference, LiteLLM/Higress model gateways, and observability. Strong in Go and Python, with Rust experience.

CORE TECHNOLOGIES

- Programming:** Go, Python, Rust
- Control Plane & Scheduling:** Kubernetes Operator, CRD / Controller, reconciliation, Service / Endpoint, finalizers, leader election, heterogeneous accelerator resource management, capacity planning and autoscaling candidates, Helm, ArgoCD
- LLM Serving & Routing:** SGLang, vLLM, prefill/decode disaggregation, KV cache, prefix routing, load-aware routing, Mooncake, RoCE/RDMA
- Gateway & Batch:** OpenAI-compatible API, Batch API, LiteLLM, Higress, Redis, MySQL, multi-tenant admission, offline concurrency budgets, shared-backend pressure control
- Reliability & Observability:** Linux, Prometheus, request tracing, QPS / TTFT / TPOT, release and rollback, failure recovery, distributed systems root cause analysis

EXPERIENCE

Zhejiang Lab	Aug 2021 - Present
Senior Research Specialist · High-Performance Computing Facility Research Center LLM Serving / Model-as-a-Service (MaaS) Platform - Own a production LLM-serving control plane built with Kubernetes Operators (CRDs/controllers), standardizing model-service declarations, worker lifecycle, Service/Endpoint readiness, P/D topology, and failure isolation for SGLang and vLLM on heterogeneous accelerators developed in China. - Designed and validated a P/D capacity-planning and autoscaling candidate based on request shapes and Capacity Profiles, completing shadow evaluation and end-to-end scaling on an isolated service with readiness gates, stabilization windows, cooldowns, and stale-intent checks. - Delivered production OpenAI-compatible multi-tenant batch inference and LiteLLM/Higress model gateways with durable sharding, concurrency budgets, and recovery, using tenant/model budgets to bound offline pressure on shared model backends. - Own releases, rollbacks, and root cause analysis for production incidents using request IDs, logs, Prometheus metrics, and Kubernetes state across model loading, queueing, gateway routing, and accelerator failures.	2024 - Present
Engineering Specialist · Intelligent Supercomputing Research Center Cloud-Native AI / HPC Platform - Customized Kubernetes Device Plugin and resource-management paths for heterogeneous accelerator discovery, node resource reporting, container-runtime validation, device-level allocation, and fault isolation. - Adapted PyTorch workloads, models, and distributed job environments to Linux distributions developed in China and heterogeneous accelerator stacks, establishing the platform foundation for later serving and scheduling work.	Aug 2021 - 2023

SELECTED ENGINEERING WORK

Candidate P/D Capacity-Planning and Autoscaling Design

Shadow · Isolated E2E · Guardrails

- Designed a candidate approach that derives request rate, input/output tokens, prefill compute/cached tokens, KV hit signals, and queue state from rolling windows, then selects Capacity Profiles by model, accelerator, quantization, TP/DP, and engine version.
- Validated independent prefill/decode demand and safe replica targets in shadow mode and through end-to-end scaling on an isolated service; added pending gates, one-step changes, stabilization windows, cooldowns, and stale-intent rejection to control execution risk.

Heterogeneous P/D Transfer and Routing

P/D Validation · RoCE/RDMA · Routing Design

- Validated the P/D functional path and high-speed network transfer configuration on accelerators developed in China: the Router selects workers and passes bootstrap/transfer context, while a transfer backend moves KV tensors over the RoCE/RDMA data plane.
- Implemented and validated routing based on request-prefix similarity and worker load, switching to least-loaded workers under imbalance and removing stale mappings when the associated worker was missing or unhealthy; evaluated Mooncake as a KV-transfer backend and designed a candidate for block routing driven by KV events.

Multi-Tenant Offline Batch Inference

Production · OpenAI Batch API · Recovery

- Used the database as the system of record, with Kubernetes Lease-based leader election, transactional claims, shard recovery, and custom_id deduplication to reduce duplicate-processing risk and verify final result completeness.
- Delivered an OpenAI Batch API-compatible workflow with streaming JSONL, tenant/model concurrency budgets, cancellation, expiry cleanup, output/error writeback, and request-level error isolation, using concurrency budgets to bound offline pressure on shared model backends.

Rust Router Priority Admission and Tracing

Independent Branch · Priority Admission · Tracing

- Implemented and tested P0-P3 admission in an independent Rust branch using global/per-priority semaphores, bounded queues, strict priority, and tenant round-robin; held permits until streaming responses completed.
- Integrated SGLang HTTP/gRPC P/D routing and bootstrap/transfer context; implemented and validated request-ID and trace-context passthrough in a dedicated branch for cross-component diagnosis.

EDUCATION

Arizona State University - M.S. in Computer Science, GPA 3.97 / 4.00, May 2019 - Dec 2020

Arizona State University - B.S. in Computer Science, Aug 2015 - May 2019

PUBLICATION & INTELLECTUAL PROPERTY

Third author, **Prompt2Length: A Lightweight Model for Predicting Output Length in LLM Inference**, IEEE AIAHPC 2025; co-inventor on two granted invention patents and named contributor on two registered software copyrights.

LANGUAGES

Mandarin Chinese (native); English (full professional proficiency).